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Beyond parallel fiber LTD: the diversity of synaptic and non-synaptic plasticity in the cerebellum

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In recent years, it has become clear that motor learning, as revealed by associative eyelid conditioning and adaptation of the vestibulo-ocular reflex, contributes to the well-established cerebellar functions of sensorimotor integration and control. Long-term depression of the parallel fiber–Purkinje cell synapse (which is often called 'cerebellar LTD') is a cellular phenomenon that has been suggested to underlie these forms of learning. However, it is clear that parallel fiber LTD, by itself, cannot account for all the properties of cerebellar motor learning. Here we review recent electrophysiological experiments that have described a rich variety of use-dependent plasticity in cerebellum, including long-term potentiation (LTP) and LTD of excitatory and inhibitory synapses, and persistent modulation of intrinsic neuronal excitability. Finally, using associative eyelid conditioning as an example, we propose some ideas about how these cellular phenomena might function and interact to endow the cerebellar circuit with particular computational and mnemonic properties.

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Cerebellar circuits and motor learning

Why bother to study the cellular substrates of memory in an atypical brain structure like the cerebellum? The simple answer is that it is the location in the mammalian brain that holds the greatest promise for uncovering the holy grail: an understanding of (admittedly simple) forms of learning and memory that will flow continuously from molecules and cells through synapses and circuits to behavior without any grossly embarrassing gaps in the middle. The factors that allow for this include an unusually well-defined circuit diagram, the fortuitous discovery of some natural cell-type specific gene promoters that will facilitate the production of transgenic mice, behavioral outputs such as eye and eyelid movements that are simple and easily quantified, and, most importantly, a clear understanding of the pathways that convey behaviorally relevant signals (such as the conditioned stimulus (CS) and unconditioned stimulus (US) in eyelid conditioning).

The cerebellar circuitry is essentially composed of a relay station in the deep cerebellar nuclei (DCN) and a cortical 'side-loop' (**Fig. 1a**). Cerebellar output to premotor centers originates in the DCN and is in turn driven by direct excitatory input from the mossy fibers. In addition, it is modulated by the inhibitory input from the Purkinje cell axons, which convey computations and interactions in the Purkinje cell. These computations will be performed upon a matrix of subtle and informationally rich excitatory parallel fiber input (~200,000 axons), massive and synchronous excitation produced by the one climbing fiber axon innervating each mature Purkinje cell, and input from inhibitory interneurons. This unusual anatomical configuration inspired a model of motor learning by Marr[1], who proposed that the parallel fiber–Purkinje cell synapses could provide contextual information, that climbing fiber–Purkinje cell synapses could signal an 'error' in motor performance that required alteration of subsequent behavior, and that the conjunction of these two signals could strengthen the parallel fiber–Purkinje cell synapse to create a memory trace for motor learning. This model was modified by Albus[2], who realized that a decrease in synaptic strength would be more appropriate given the sign-reversing function of the Purkinje cell inhibitory output. Albus also noted that this model is analogous to classical conditioning with the parallel fibers conveying a CS, the climbing fiber a US and a depression of the parallel fiber–Purkinje cell synapse giving rise to a conditioned response (CR) via disinhibition of the DCN. To place this model in a behavioral context, let us consider associative eyelid conditioning (**Fig. 1b**). Before training, an airpuff to the eye (US) gives rise to an immediate reflexive blink (the unconditioned response, UR). During training, a neutral stimulus such as a tone (CS) is paired with the airpuff stimulation so that the tone onset precedes the airpuff and the two stimuli coterminate. As the association is acquired, a carefully timed eyelid response is performed (the CR), the onset of which shortly precedes the airpuff. This learning can also be actively reversed. In well-trained animals that reliably perform CRs, this response can undergo rapid 'extinction' if tone (CS) stimuli are repeatedly presented without airpuffs.

There is extensive evidence to support the involvement of cerebellar cortical and deep nuclear circuits in associative eyelid conditioning (for review, see refs. 3 and 4). First, extracellular recording showed that in well-trained animals, populations of cells in the DCN begin to fire during the CS–US interval (**Fig. 1b**). This firing is predictive of and correlated with the performance of the CR, suggesting that the CR behavior is expressed

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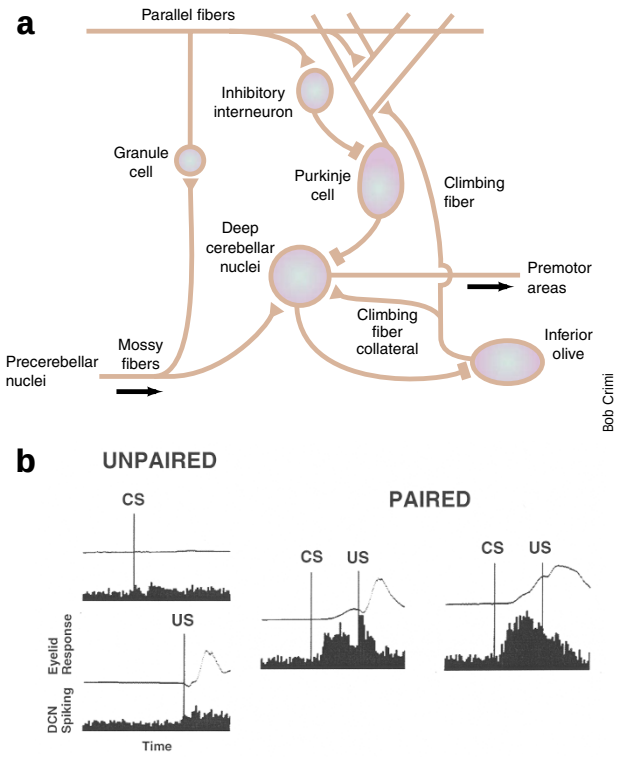

Fig. 1. Cerebellar circuitry. (**a**) The neurons of the deep cerebellar nuclei (DCN) receive their main excitatory input from glutamatergic mossy fibers (MFs), which are the axons of a large number of pre-cerebellar nuclei. The main outflow of information from this structure is carried by excitatory axons that originate from the large neurons of the DCN and project to premotor areas and thalamic structures. The sole output of the cortical side-loop is the inhibitory, GABAergic projection from Purkinje cells to the neurons of the DCN. Purkinje cells receive two major excitatory inputs, which are organized in very different ways. Each mature Purkinje cell is innervated by a single glutamatergic climbing fiber, which is the axon of cells in the inferior olive. This is potentially the most powerful synaptic contact in the brain, as a each Purkinje cell receives ~1,400 synapses from a single climbing fiber axon. Climbing fibers also provide a weak collateral innervation of the DCN consisting of a few synapses in the distal dendrites, the function of which is poorly understood. In contrast, each Purkinje cell receives ~200,000 synapses from parallel fibers, which are the axons of granule cells. Parallel fibers also drive GABAergic inhibitory interneurons (stellate and basket cells), which project to the axo-somatic regions of Purkinje cells. Closing the loop, granule cells receive excitatory synapses from branches of the same mossy fibers that innervate the DCN directly. Excitatory presynaptic terminals are drawn as triangles, and inhibitory presynaptic terminals are drawn as bars. (**b**) Associative eyelid conditioning. In each panel, the upper trace represents the eyelid response and the lower histogram represents extracellularly recorded spiking in the DCN. No response is seen to the CS and a reflexive UR is seen to the US (CS and US onset are indicated by vertical lines. With repeated pairing of the CS and US, a blink response develops that is timed to precede the US onset. This CR is correlated with an increase in the firing rate of DCN cells. This panel is modified from ref. 75. Copyright 1984 by the Society for Neuroscience.

in the firing rate and pattern of DCN neurons. Furthermore, stimulation of the DCN can elicit eyelid responses in naive animals. Second, during training, stimulation of mossy and climbing fibers can substitute for the CS and US, respectively. Finally, reversible inactivation of the DCN and overlying cerebellar cortex prevented the acquisition of the eyelid CR, but not the performance of the UR. In contrast, inactivation of the superior cerebellar peduncle or red nucleus, sites through which excitatory DCN output is conveyed, prevented the expression of the CR during training, but not its acquisition, as evidenced by the fact that the CR was present after inactivation. These studies suggest that the cerebellum and its associated projections are essential for acquisition and expression of the eyelid CR. More specifically, the memory trace seems to be localized upstream of the red nucleus but downstream of the mossy and climbing fibers, that is, in the cerebellar cortex and/or the DCN. Similar evidence implicates the cerebellar cortex and vestibular nuclei in another form of motor learning, adaptation of the vestibulo-ocular reflex[5] (reviewed in ref. 6). Although it is likely that there are other forms of motor learning that require plasticity in the cerebellar circuit (multijoint limb load adjustment is a good candidate), it should be cautioned that there are also forms of motor learning (such as improvement in the performance of rotating rod and thin rod balancing tasks) that can persist even when some forms of cerebellar plasticity are blocked[5].

Parallel fiber LTD

Cellular mechanisms that might underlie acquisition of associative eyelid conditioning would be expected to be located at points in the cerebellar circuit where CS and US signals converge and where their repeated co-occurrence results in an increase in DCN firing. As suggested by Marr–Albus models, parallel fiber LTD is one such cellular mechanism. In parallel fiber LTD, first reported by Ito and collaborators, a persistent attenuation of the parallel fiber–Purkinje cell synapse is produced when parallel fiber and climbing fiber inputs to a Purkinje cell are stimulated together at low frequency[7]. The requirements for parallel fiber LTD induction, like those for eyelid conditioning, are typically associative—parallel fiber synaptic strength is not attenuated by climbing fiber stimulation alone, parallel fiber stimulation alone, or stimulation of both in an unpaired manner. (See refs. 8 and 9 for examples of LTD induced with very strong PF stimulation alone.) Repeated induction of parallel fiber LTD results in saturation of the response; a maximum depression of ~50% is typical. Parallel fiber LTD is generally understood to have three initial requirements for induction. The climbing fiber contributes to LTD induction via Ca^{2+} influx through voltage-gated channels, occurring during the complex spike. The parallel fibers release glutamate, which acts upon both mGluR1 metabotropic receptors and AMPA receptors. After these initial signals, there is a requirement for transient protein kinase C (PKC) activation. The involvement of a nitric oxide/cGMP/protein kinase G/phosphatase inhibition pathway has also been suggested. (For review, see ref. 10.) Parallel fiber LTD is expressed postsynaptically, as a reduction in the number of functional AMPA receptors produced by clathrin-mediated endocytosis[11,12].

Parallel fiber LTP

In an animal that has been trained to achieve asymptotic performance of eyelid CRs, one way in which repeated application of the CS alone could be made to result in CR extinction would be to produce homosynaptic LTP of parallel fiber–Purkinje cell synapses. Indeed, this form of LTP may be observed after low-frequency (2–8 Hz) parallel fiber stimulation in the absence of climbing fiber activation[13–18]. In contrast to LTD, LTP at the parallel fiber synapse seems to be triggered by a presynaptic Ca^{2+} influx. Its induction is blocked by removal of external Ca^{2+}, but not by antagonists of glutamate receptors or by loading of the Purkinje cell with the

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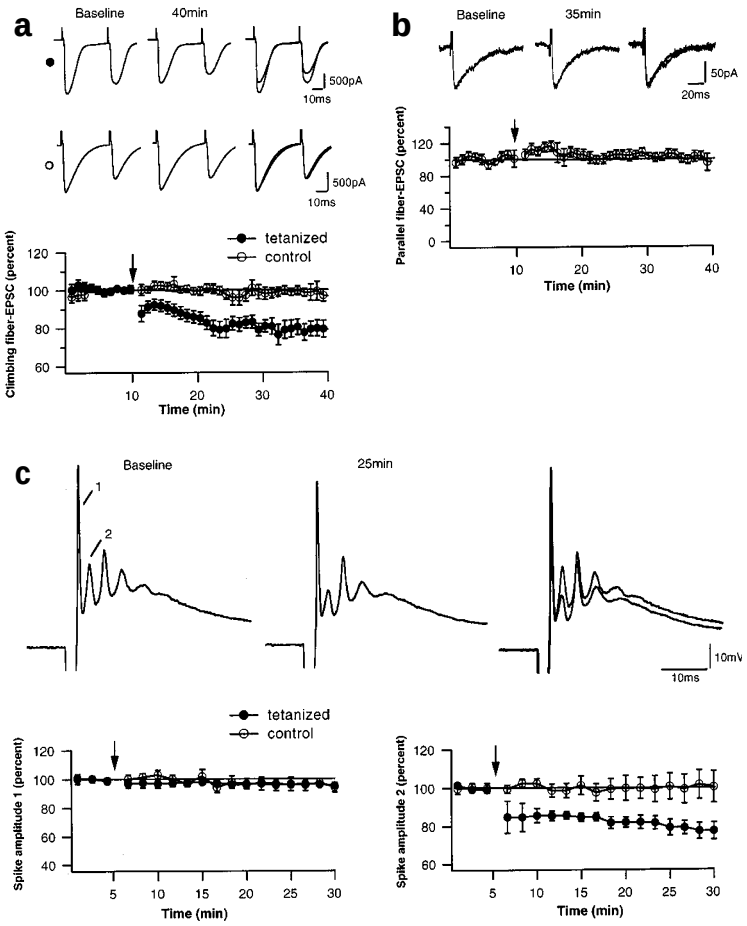

Fig. 2. Climbing fiber long-term depression. (**a**) 5 Hz × 30 s climbing fiber stimulation resulted in a depression of climbing fiber-evoked EPSCs (n = 15). Under control conditions, no change in EPSC amplitude was observed (n = 5). Each test stimulus consisted of a pulse pair applied with an interval of 50 ms. Traces are averages of two to five sweeps. In the time-course graph, each data point represents the average of three successive responses evoked at 0.05 Hz. Only the amplitude of the first EPSC is shown. (**b**) Climbing fiber tetanization did not change the amplitudes of parallel fiber EPSCs as measured in a subset of the tetanized cells (n = 9) shown in (**a**). (**c**) Climbing fiber stimulation produced an alteration in the complex spike waveform (current-clamp mode). Representative single traces are shown before and 25 min after tetanization. The time-course graphs show the changes in the amplitudes of the first two spike components under tetanization (n = 5) and control conditions (n = 4). Each data point represents the average of five successive responses evoked at 0.05 Hz. This figure is modified from ref. 29. Used with permission from the copyright holder, Cell Press.

Ca^{2+} chelator BAPTA[19–22]. Several lines of evidence suggest that parallel fiber LTP is mediated by the presynaptic activation of Ca^{2+}-sensitive adenylyl cyclase, leading to a rise in the concentration of cAMP and the consequent activation of protein kinase A (PKA). An LTP-like effect may be produced by bath application of the adenylyl cyclase activator forskolin or membrane-permeable cAMP analogs[19,23–25]. In addition, cerebellar LTP induced by granule cell stimulation (but not an exogenous cAMP analog) is attenuated in cell cultures from a type I Ca^{2+}-sensitive adenylyl cyclase null mutant mouse[25]. Induction of LTP produced by tetanic stimulation is blocked when PKA inhibitors are applied in the bath[19,24] or when presynaptic granule cells are transfected with an expression vector encoding a peptide inhibitor of PKA[22].

The expression of parallel fiber LTP also seems to be presynaptic. It is associated with a decrease in the rate of synaptic failures[16,20,21,25] and the extent of paired-pulse facilitation[19,21]. Further evidence comes from the observation that LTP could be detected in granule cell–glial cell pairs in culture when postsynaptically recording evoked AMPA/kainate receptor-mediated currents[20] or glutamate transport currents[21]. The observation that such 'reporter currents' measured in glial cells can also undergo LTP indicates a presynaptic locus of expression. Thus, whereas LTD at the parallel fiber–Purkinje cell synapse is postsynaptically induced and expressed, LTP is presynaptically induced and expressed. In such a scheme, cerebellar LTP and LTD would not truly reverse each other, but rather would be additive, independent phenomena, a suggestion that has recently been confirmed experimentally[10]. This is in striking contrast to results obtained at excitatory synapses received by pyramidal cells in the hippocampal CA1 subfield, in which both LTP and LTD can be postsynaptically expressed and the synapses are endowed with reversible bidirectional modification (for review, see ref. 10).

Climbing fiber LTD

Stimulation of the one climbing fiber axon that innervates a mature Purkinje cell results in the activation of ~1,400 synapses with a high release probability[26], thereby reliably producing a massive excitation of the Purkinje cell known as the complex spike. The classic Marr–Albus–Ito models of cerebellar function are generally silent regarding changes in the strength of this climbing fiber–Purkinje cell synapse. However, climbing fibers show a form of developmental plasticity during the first three weeks of postnatal life. After an initial innervation of each Purkinje cell by several climbing fibers, an activity-dependent elimination of surplus climbing fibers takes place until the 1:1 connection ratio that is typical for the adult cerebellar cortex is achieved[27]. Moreover, even in adult rats, climbing fibers show some degree of morphological plasticity, depending on the availability of target Purkinje cells and the overall electrical activity of the cerebellar cortex[28]. Furthermore, the molecular machinery required for LTD at the parallel fiber synapse (mGluR1, P/Q-type Ca^{2+} channels, GluR2-containing AMPA receptors and PKC) is also present at the climbing fiber input.

These observations inspired a study using whole-cell patch-clamp recordings in a slice preparation in which climbing fiber LTD was observed and characterized for the first time[29]. LTD of climbing fiber-evoked EPSCs was observed after 5 Hz × 30 s tetanization (**Fig. 2**). This climbing fiber LTD did not spread to neighboring parallel fiber synapses, and, like parallel fiber LTD, it could be saturated when tetanization was repeated. LTD was not associated with a change in the paired-pulse ratio. This latter result is preliminary evidence for a postsynaptic locus of expression (assuming that a presynaptic modification would change the release probability). Similar to parallel fiber LTD, climbing fiber LTD could be blocked by loading the Purkinje cells with the Ca^{2+} chelator BAPTA, or by bath application of a group I mGluR antagonist or a PKC inhibitor.

What might be the consequences of climbing fiber LTD for cerebellar function? Perhaps, because of a high safety factor for

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evoking complex spikes, the ~20% attenuation of EPSC amplitude produced by climbing fiber LTD is of no relevance for cerebellar function and is merely an 'evolutionary byproduct' of the biochemical mechanisms used for parallel fiber LTD. However, when recordings were performed in current-clamp mode, 5 Hz × 30 s stimulation resulted in a selective modification of the complex spike. Complex spikes are composed of an initial, somatically mediated Na^+ spike, followed by a series of smaller spikes that are produced by dendritic Ca^{2+} influx[30,31] and a recently described 'resurgent Na^+ current'[32]. After 5 Hz tetanization, the second complex spike component was selectively reduced, suggesting that climbing fiber LTD can indeed influence the integrative response of the Purkinje cell. Complex spikes are associated with a large dendritic Ca^{2+} transient[33,34], which is likely to be attenuated as a consequence of climbing fiber LTD. This Ca^{2+} transient can have several important effects. It can briefly modulate dendritic excitability through effects on ion channels such as Ca^{2+}-dependent K^+ channels as well as other electrogenic proteins such as the Na^+/Ca^{2+} exchanger. In addition, climbing fiber-evoked Ca^{2+} transients are critical for induction of parallel fiber LTD[14,34], short-term potentiation of mGluR-mediated responses at the parallel fiber synapses[35], and both LTP[36] and transient depression[37] of GABAergic interneuron–Purkinje cell synapses.

LTP of GABAergic interneuron–Purkinje cell synapses

LTP of both spontaneous and evoked $GABA_A$ receptor-mediated IPSCs may be induced by repetitive climbing fiber activation[36]. This inhibitory LTP (called 'rebound potentiation') requires a postsynaptic Ca^{2+} transient, as it can be blocked by loading Purkinje cells with BAPTA and can be induced after application of depolarizing voltage steps in place of climbing fiber stimulation. The Ca^{2+} transient that triggers rebound potentiation seems to require Ca^{2+} mobilization from internal stores[38,39]. Inhibitors of either calcium/calmodulin-dependent kinase II (CaMKII)[40] or PKA[41] have been reported to block this form of LTP. The role of inhibitory synaptic activity in this process is somewhat controversial. It has been suggested that simultaneous activity of inhibitory synapses is needed for the induction of rebound potentiation[42]. This finding is in contrast to the recent demonstration that simultaneous inhibitory activity can suppress rebound potentiation when inhibitory interneurons are activated within ~1 s of climbing fibers. This effect was described to be mediated through a cascade involving postsynaptic $GABA_B$ receptors and inhibition of adenylyl cyclase and cAMP production[41]. If the latter finding prevails, it would comprise an unusual computation: a heterosynaptic potentiation that is suppressed by homosynaptic synchrony. A protocol for reversing this form of LTP has yet to be described.

Alterations of interneuron–Purkinje cell synaptic strength are likely to have a major influence on Purkinje cell throughput. Indeed, it has been demonstrated that single action potentials evoked in inhibitory interneurons can generate delays in Purkinje cell action potential firing and that tonic inhibitory input can drastically modulate the spike firing pattern in Purkinje cells[43].

LTP of mossy fiber–granule cell synapses

Mossy fibers constitute a large glutamatergic system of the brain and have granule cells as their primary target (there are as many as 10^{11} granule cells and 4 times as many mossy fiber–granule cell synapses), providing a large potential substrate for information storage. Activation of mossy fibers using either a theta burst pattern (100 Hz for 100 ms repeated 8 times at 250 ms intervals), or a single tetanus (100 Hz for 1 s) that mimics endogenous activity[44,45], paired with postsynaptic depolarization to –40 mV, results in LTP of mossy fiber EPSCs[46,47] (**Fig. 3d** and **e**), whereas protracted low-frequency stimulation (2 Hz for 5 min) causes LTD (Maffei, Rossi and D'Angelo, unpublished data). LTP is similar to that commonly recorded in hippocampal area CA1 in that it requires postsynaptic depolarization, NMDA receptor activation and consequent Ca^{2+} influx. As such, it is quite sensitive to the level of inhibitory input[47], which, in this case, is provided by Golgi cells[48]. Mossy fiber LTP can also be blocked by an mGluR antagonist or a PKC inhibitor[46,49]. During mossy fiber LTP expression, amplitudes of both the AMPA and the NMDA receptor-mediated synaptic currents are increased. Mossy fiber LTP expression is also associated with a change in the kinetics of the NMDA-EPSC such that deactivation is protracted. This could constitute an extension of the window for EPSP temporal summation and, thereby, coincidence detection. The role of mossy fiber–granule cell LTP/LTD is not completely understood. Whereas LTP would promote population coding, LTD would promote sparse coding in the cerebellum granular layer[1,2]. A regulation of synaptic strength at this synapse, together with changes in intrinsic excitability and in the strength of Golgi cell–granule cell connections, may have a critical influence on the selection of mossy fiber patterns to be relayed to the cerebellar cortex[50].

LTP and LTD of synapses received by DCN cells

Although LTP and LTD of synapses that are received by granule or Purkinje cells are one possible way to achieve acquisition and extinction of eyelid conditioning, plasticity of the synapses received by the DCN cells could also modulate DCN firing and hence CR expression. Unfortunately, only a single preliminary report of use-dependent plasticity of mossy fiber–DCN synapses has appeared in the literature. Racine and collaborators[51] used field potential recording in both acute and chronic preparations of intact rat cerebellum to show that a component of the field potential was potentiated by repeated bursts of 300 Hz stimuli to the inferior peduncle. This potentiation ranged from 118–150% of baseline, was seen in 5 of 6 animals and lasted up to 8 days in chronic recordings.

Recent work has indicated that the GABAergic Purkinje cell–DCN synapses also exhibit LTP and LTD[52]. Activation of these synapses with a burst-and-pause stimulus results in a train of summating hyperpolarizing IPSPs followed by a prominent rebound depolarization and associated spike burst. This rebound depolarization provides a mechanism by which inhibitory inputs can drive postsynaptic excitation. In these cells, LTP is elicited by trains of IPSPs, which reliably evoke a rebound depolarization in the DCN neurons. LTD is induced if the same protocol is applied while the amount of postsynaptic excitation is reduced (by postsynaptic hyperpolarization or an internal Na^+ channel blocker), and no change is produced if a Ca^{2+} chelator is applied postsynaptically. The polarity of the change in synaptic strength is correlated with the amount of rebound depolarization-evoked spike firing and the amplitude of the resulting postsynaptic Ca^{2+} transient. These results speak to two important points that have implications beyond cerebellar function. First, they suggest a solution for a long-standing problem in cellular synaptic plasticity: how do inhibitory synapses produce postsynaptic Ca^{2+} signals that are required for use-dependent plasticity when they do not directly gate a Ca^{2+} conductance? Second, they show that an important computational principle that governs the induction of use-dependent change in excitatory synaptic efficacy, namely that LTP and LTD are triggered by stimuli that result in high and moderate Ca^{2+} signal amplitudes, respectively[53–55], can also apply to inhibitory synapses. It remains to be determined whether the

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LTP and/or LTD thresholds can also be changed as a function of recent synaptic activity (so-called 'metaplasticity,' another prediction of the excitatory synapse models cited above).

It is likely that any stimulus that appropriately elevates subsynaptic Ca^{2+} in DCN neurons will result in LTP/LTD of Purkinje cell–DCN synapses. Indeed, LTP may be produced either by repeated postsynaptic hyperpolarizing pulses, which evoke rebound depolarization[52] or by depolarizing pulses[56]. The expression mechanism remains unclear, as it could be detected using exogenous pulses of a $GABA_A$ receptor agonist, suggesting a postsynaptic locus of expression, yet it was associated with an increase in the frequency but not the amplitude of mIPSCs, suggesting a presynaptic locus of expression[56]. LTD of inhibitory input to the DCN may also be produced using postsynaptic depolarizing pulses delivered at a lower frequency[57]. Both LTP and LTD of inhibitory input may be induced heterosynaptically, by repeated stimulation of glutamatergic mossy fiber synapses at 100 or 10 Hz, respectively[56,57]. Both LTP and LTD of these inhibitory synapses induced in this manner require NMDA receptor activation and elevation of postsynaptic Ca^{2+}. At present, it is unclear whether induction of LTP/LTD in an NMDA receptor-independent manner[52] and an NMDA receptor-dependent manner[56,57] engage the same signaling cascades downstream of postsynaptic Ca^{2+} elevation.

Non-synaptic plasticity in cerebellar neurons

Memory storage in the brain is generally assumed to be mediated by long-term modifications in the strength of synaptic transmission. Whereas LTP and LTD are computationally appealing (in part because they allow for the synapse-specific modification of a large array of inputs), and it is likely that they are central to memory storage, there are also reasons to believe that they are not the whole story. One idea, which has received relatively little attention, is that information storage in the mammalian brain may also involve activity-dependent changes in the intrinsic excitability of neurons. This type of plasticity could be a local phenomenon that is restricted to one synapse, but it could also have the property of affecting postsynaptic signals evoked by a large number of (or possibly even all) synapses impinging on a particular neuron, depending upon the identity and location of the intrinsic (voltage-sensitive) conductances altered, thereby creating a more global change in signal integration. Until recently, there have only been scattered results in support of a role for intrinsic, non-synaptic plasticity in memory storage. For example, there is evidence for persistent changes in intrinsic neuronal excitability induced by behavioral training. Studies using *Hermissenda* have shown that associative pairing of light and rotation produces a long-term increase in the intrinsic excitability of type B photoreceptors, due to a reduction of K^+ currents[58]. A similar effect has been seen following various forms of associative conditioning in neocortical neurons[59], CA1 pyramidal neurons of the hippocampus[60] and cerebellar Purkinje cells[61]. In addition, slowly developing changes in the intrinsic excitability of both invertebrate and mammalian neurons have been reported in cell culture systems when activity was blocked with tetrodotoxin[62,63]. However, rapid, synaptically driven changes in intrinsic excitability of mammalian neurons had not been convincingly demonstrated.

Aizenman and Linden[64] made current-clamp recordings from the DCN of juvenile rat cerebellar slices in which a short depolarizing test pulse that evoked a small number of action potentials was applied. After a baseline period, a synaptic tetanus was

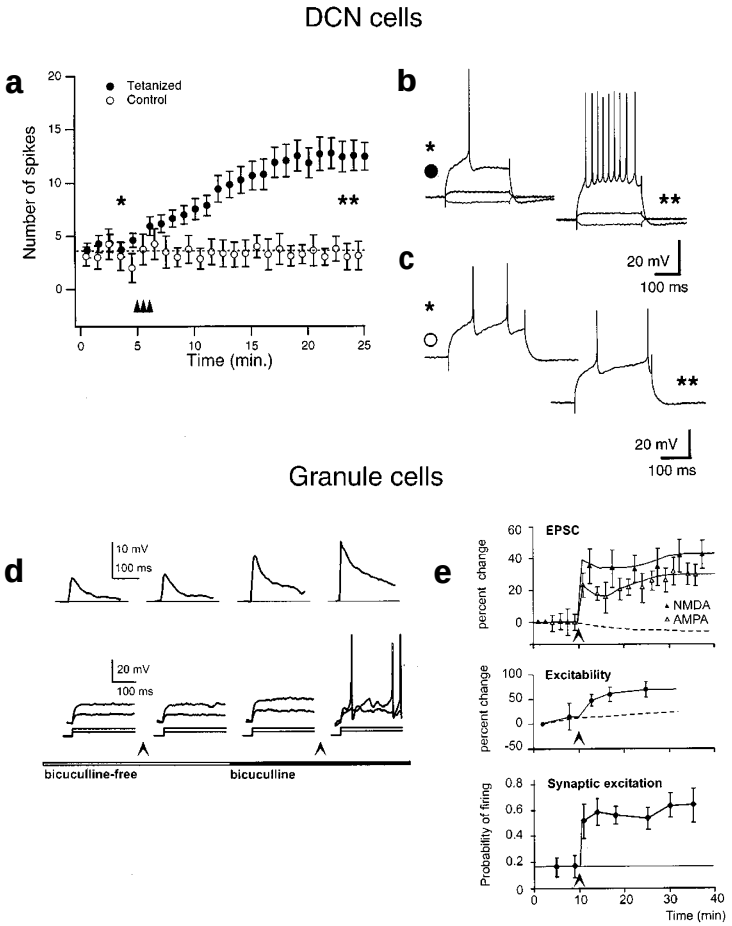

Fig. 3. Rapid, synaptically driven increases in intrinsic excitability of DCN and granule cells. (**a**) Tetanization of excitatory synapses produced a sustained increase in intrinsic excitability of DCN neurons. Averaged time course of the number of spikes evoked during 200 ms, depolarizing current test pulses. Control experiments (open circles) show that a stable baseline can be recorded for at least 25 min. Application of a synaptic conditioning tetanus (at arrowheads, filled circles) resulted in a slowly developing increase in the number of spikes evoked by the test stimulus. Error bars, s.e.m. Representative traces, evoked by a +0.5 nA test pulse, from tetanized (**b**) and control (**c**) experiments at the time points indicated by asterisks. Traces evoked by sub-threshold test pulses demonstrate that R_{input} remains stable when an increase in spiking is observed with larger depolarizing test pulses. (**d**) Tetanization of mossy fibers (arrowheads) produced a sustained increase in synaptic responses and intrinsic excitability of granule cells when synaptic inhibition was turned off (bicuculline) but not when it was left active (bicuculline-free). The nature of synaptic and non-synaptic changes is illustrated in (**e**). EPSCs (comprising both AMPA and NMDA components) increased following tetanization (arrowhead). Membrane excitability (the opposite of current threshold in current-clamp recordings) also increased. The combination of these changes caused a large increase in synaptic excitation, evaluated as the probability of action potential generation by EPSPs. Dotted lines in the top and middle panel are interpolation of control experiments, showing stable baselines throughout LTP recordings. (**a–c**) From ref. 64. (**d**, **e**) From refs. 46 and 47. Copyright 1999 and 2000 by the American Physiological Society and the Society for Neuroscience, respectively.

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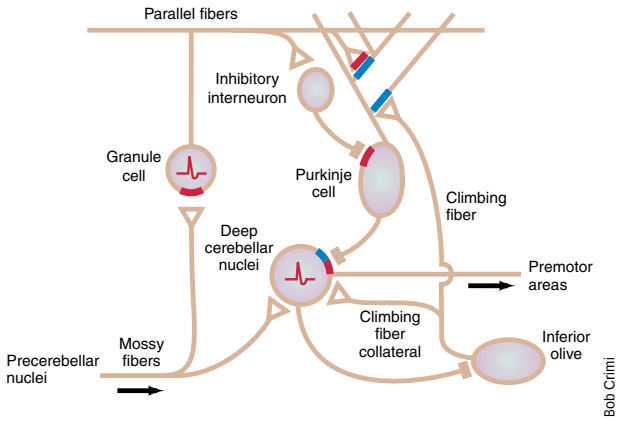

Fig. 4. A summary of synaptic and non-synaptic plasticity in the cerebellar circuit. In this drawing, the occurrence of long-term use-dependent plasticity has been coded with color: red indicating potentiation and blue indicating depression. Red action potentials in neuronal somata indicate persistent increases in intrinsic excitability, whereas bars of color at synapses indicate conventional synaptic LTP or LTD. The location of the color bar in the presynaptic or postsynaptic membrane represents the best current understanding of the locus of expression of these forms of LTD. In some locations, such as the PF–PC synapse, the data in support of these expression sites are good, whereas in others, such as the PC–DCN synapse, much remains to be determined. It is likely that additional forms of long-term synaptic and non-synaptic plasticity will have to be added to this diagram in the near future.

delivered to the DCN neuron using a stimulating electrode placed in adjacent white matter, to activate glutamatergic mossy fibers. ($GABA_A$ receptors were blocked with picrotoxin.) This synaptic conditioning tetanus resulted in a slowly developing but dramatic increase in the number of spikes evoked by the test pulse, which stabilized over a period of 15–20 min and which was associated with a decrease in the spike threshold (**Fig. 3a–c**). Subsequent experiments showed that induction of this persistent increase in intrinsic excitability required Ca^{2+} influx mediated by either NMDA receptors (when triggered by a synaptic conditioning tetanus) or voltage-sensitive Ca^{2+} channels (when triggered by repeated, direct somatic current injection). Thus, whereas these studies showed this persistent increase in intrinsic excitability to be driven by bursts delivered to the mossy fibers, it is likely that any stimulus that results in the appropriate postsynaptic Ca^{2+} transient will produce this effect. For example, a burst-and-pause stimulus delivered to the GABAergic Purkinje cell–DCN synapses results in rebound depolarization and an associated Ca^{2+} transient[52,65], which would be an excellent candidate for such a signal.

When theta burst stimulation (bursts of 10 pulses with an interpulse interval of 10 ms, repeated with an interburst interval of 200 ms, designed to mimic firing during theta oscillations) plus postsynaptic depolarization was applied to mossy fiber–granule cell synapses, a persistent increase in the intrinsic excitability of the postsynaptic granule cell was observed[47] (**Fig. 3d** and **e**). As previously described, this form of stimulation also gives rise to LTP of the mossy fiber EPSCs. Like mossy fiber–granule cell LTP, the increased excitability was dependent upon postsynaptic depolarization, NMDA receptor activation and a postsynaptic Ca^{2+} transient. However, the increase in intrinsic excitability required less membrane depolarization ($\geq$ 60 mV) than the mossy fiber LTP ($\geq$ 40 mV). Thus, weak granule cell depolarization shall only induce the increase in membrane excitability, whereas strong granule cell depolarization shall increase membrane excitability and synaptic conductance together. Once again, the influence of membrane depolarization suggests that Golgi cell inhibition is critical for fine tuning plasticity at the mossy fiber–granule cell relay. Potentiation of intrinsic excitability may be relevant in conditions of low synaptic excitation, restoring granule cell readiness (compensatory effect[66]) and regulating the number of granule cells effectively activated by discharging mossy fibers. At present, it has yet to be determined if the use-dependent increases in intrinsic excitability in granule or DCN cells may be actively reversed.

Use-dependent alterations in intrinsic excitability, as shown in cerebellar granule and DCN cells, have the potential to broadly affect neuronal throughput. Indeed, the probability of spike firing as evoked by every excitatory synapse on a postsynaptic neuron may be altered. This is based on the assumption that either the postsynaptic neuron is extremely electrotonically compact (as is the case for the cerebellar granule cell[67]) or, in noncompact neurons, the conductances that are persistently altered by depolarizing current injection are physically interposed between excitatory synapses and the spike initiation zone, as would be expected if they were localized to the soma or axon hillock. The spatial localization of these conductances has yet to be determined and may be of great computational importance. For example, if a group of synapses impinging upon a DCN neuron were activated to evoke a local Ca^{2+} transient, this might produce a local change of voltage-gated ion channels and thereby modulate throughput, not from individual synapses, or from a whole neuron, but rather from a dendritic module consisting of the originally activated synapses plus all those more distal on the same dendritic branch.

Emergent properties of a plastic cerebellar circuit

Let us return to the case of associative eyelid conditioning. Can the cerebellar circuit, with the complement of use-dependent mechanisms described herein (**Fig. 4**), be reasonably expected to underlie this simple form of learning and memory? To account for acquisition of the CR, we need a location or locations in the circuit where CS and US information converge and where an associative computation is performed that ultimately results in an increase in firing in the DCN. This increase in firing must be appropriately timed to shortly precede the US onset. Furthermore, to account for extinction of the CR, we need a location or locations in the circuit where CS information is received and where its repeated presentation alone will ultimately result in a decrease in DCN firing during the CS–US interval. In addition to acquisition and extinction, it would be useful if the cerebellar circuit could account for some of the more complex phenomena related to associative eyelid conditioning. For example, if an animal is repeatedly presented with a tone (CS)–airpuff (US) pairing until it reliably acquires tone-evoked eyelid responses (CRs), but is then presented with repeated tones alone, it will undergo 'extinction' so that it no longer performs eyelid responses to the tone. However, if tone–airpuff pairing is then resumed, the rate of acquiring the tone-evoked eyelid response will be greater in this second training session. This is referred to as 'savings.' A related phenomenon is called 'CS generalization.' This refers to the finding that an animal that has previously undergone acquisition and complete extinction to a CS–US pair (tone–airpuff pairs followed by repeated tones presented alone), in a second training session, will acquire eyelid response CRs faster to pairing of the airpuff US with a different CS, such as a light. Finally, 'US pretreatment' refers to the observation that repeated presentations of a US before CS–US pairing will result in slower CR acquisition, as if the US were somehow less effective.

Clearly, one place where the streams of CS and US information converge is the Purkinje cell, in which parallel fiber LTD

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could drive CR acquisition through DCN disinhibition as envisioned in the models of Marr, Albus, Ito and Thompson. How can this mechanism produce CRs that are appropriately timed (so that the CR begins shortly before the US onset)? If one assumes that a large population of granule cells codes for the CS in a continuum of different ways, with some firing at the beginning of the CS, others at the end of the CS, and others in the middle, then those PF synapses from granule cells that fire in close proximity to US onset will be depressed, whereas those that fire far from US onset will be potentiated by parallel fiber LTP. When transduced by the Purkinje cell–DCN inhibitory synapse, this will result in lower DCN firing rates in the period immediately following CS onset and higher firing rates shortly before US onset, thereby producing a well-timed CR[68,69].

However, the Purkinje cell is not the only site that receives convergent CS and US information. The DCN cells receive both mossy fiber and climbing fiber collateral synapses. Unfortunately, the latter have not been well characterized physiologically. If climbing fiber/mossy fiber coactivation resulted in either LTP of mossy fiber DCN synapses[51] or a persistent increase in DCN intrinsic excitability[64], this could potentially underlie at least a portion of CR acquisition (see below). The main route for climbing fibers to excite the DCN may be disynaptic. When climbing fibers fire, there is a transient increase in the firing rate of Purkinje cells, followed by a pause. When this pattern of activity is applied to Purkinje cell–DCN synapses, a strong rebound depolarization and consequent Ca^{2+} transient is produced[52,65].

We must also consider that the memory trace for CR acquisition may be stored in multiple cerebellar locations and that plasticity at one site may induce plasticity at another site as a secondary consequence. For example, in a model proposed by Mauk and collaborators, the memory trace of the eyelid CR is sequentially stored, first as a depression of the parallel fiber–Purkinje cell synapse in the cerebellar cortex. This would result in an attenuation of Purkinje cell firing and hence Purkinje cell–DCN synaptic input, thereby disinhibiting the DCN targets. This disinhibition, when coupled with activation of the mossy fiber–DCN synapse, could then potentiate the latter or increase the DCN intrinsic excitability, resulting in a distributed memory trace. In this scheme, aspects of the CR (mostly amplitude) are stored at the mossy fiber–DCN synapse, whereas other aspects of the CR (mostly timing information) are retained in the parallel fiber Purkinje cell synapse[68,70,71]. A computational analysis has suggested that this model could constitute a memory trace that is unusually resistant to degradation by ongoing 'background' activity in the cerebellar circuit[72]. Furthermore, granule cell populations active at specific times may be consolidated by LTP or LTD at the mossy fiber–granule cell synapse, or by changes in granule cell intrinsic excitability.

There are several ways in which repeated CS information could ultimately give rise to decreased DCN firing and consequent extinction of CRs. Both the mossy fiber–granule cell synapse and the parallel fiber–Purkinje cell synapse undergo LTP with repeated activation. In addition, granule cell intrinsic excitability is increased following repeated mossy fiber stimulation. If the acquisition of the CR involves a particular modification at a specific synaptic or cellular location, should extinction produce a complete and exact reversal of these effects? Whereas LTD and LTP of the parallel fiber synapse are obvious candidates to underlie CR acquisition and extinction, respectively, they do not truly reverse each other, as the former is expressed postsynaptically, whereas the latter is expressed presynaptically. Although a parallel fiber synapse may undergo LTD, and then LTP, to return it to the same aggregate synaptic strength as measured with single pulses, this synapse will not then function in the same way as it did in the naive state. For example, if it now has a higher probability of release, it will be more subject to fatigue during prolonged firing. This situation may become even more complex if a portion of the memory trace for CR acquisition (such as the mossy fiber synaptic strength or intrinsic excitability in the DCN) is not altered at all following extinction.

Lest we become too discouraged here, it should be noted that the 'reversibility problem' may have a silver lining. Clearly, to account for savings and CS generalization, the state of the cerebellar circuit after CR acquisition followed by extinction cannot be identical to that before training. Residual plasticity in one or more cerebellar locations is likely to underlie these phenomena. Recently, local reversible inactivation during training, combined with a computer simulation, has suggested that the deep nuclei might be one site of this residual plasticity[71]. Whereas this residual plasticity could take the form of either LTP of the mossy fiber–DCN synapse or an increase in intrinsic excitability, the latter may provide a better explanation for generalization between CSs that are coded by distinct populations of mossy fibers.

In some forms of associative classical conditioning, repeated US presentation before CS–US pairing slows the rate of CR acquisition. Presumably, it is less behaviorally relevant to learn a CS–US association if the US alone has been previously applied repeatedly. Repeated US presentation will activate climbing fibers and may result in LTD of climbing fiber–Purkinje cell synapses. This could then slow down acquisition of CRs in two ways, both related to reduction of the climbing fiber-evoked Ca^{2+} transient. First, a smaller Ca^{2+} transient could make it harder to induce parallel fiber-Purkinje cell LTD from subsequent parallel fiber/climbing fiber coactivation. Second, a smaller Ca^{2+} transient could reduce the amplitude of the Ca^{2+}-dependent K^+ conductance, which is a good candidate for causing the climbing fiber-evoked pause in ongoing Purkinje cell spiking. This could then have a secondary effect of delimiting rebound excitation in the DCN and thereby attenuating the potential contribution of climbing fiber activation to synaptic and non-synaptic plasticity in this structure.

To this point, we have made a set of proposals about how some cellular phenomena in the cerebellum might constitute the memory trace for a simple form of motor learning and how their interaction might account for some known properties of this learning such as extinction, savings, CS generalization and so forth. Although this might seem to be sufficient, it is likely that this circuit must have the capacity to simultaneously encode more than this one simple form of motor learning. Thus, in addition to the cellular alterations that initially lay down the memory trace, there must also be a set of homeostatic phenomena that regulate the overall activity of the circuit so that the efficacy of synapses or the intrinsic properties of neurons are not driven either to their absolute minimum or maximum[63,73,74]. The latter is particularly dangerous in the case of excitatory synapses, because not only is the subsequent information storage capacity of the network compromised, but there is also the risk of damage by Ca^{2+}-mediated excitotoxic processes.

There are several phenomena in the cerebellar circuit that may serve homeostatic or antiexcitotoxic functions. For example, the climbing fiber input evokes massive excitation of the Purkinje cell, which results in large dendritic Ca^{2+} transients[33,34]. At the same time, bursts of climbing fiber activity may provide a means for general activity reduction in the cerebellar cortex in that it can induce LTP of inhibitory interneuron–Purkinje cell synapses and LTD of coactive excitatory parallel fiber synapses, as

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well as homosynaptic climbing fiber LTD. Similarly, LTP and LTD of Purkinje cell–DCN synapses may serve as a homeostatic mechanism to regulate the overall excitability and dynamic range of the DCN cell. For example, the increase in intrinsic excitability that can be produced by strong postsynaptic spiking[64] could constitute a positive feedback loop that would drive spiking to dangerously high levels. However, large amounts of postsynaptic spiking will also potentiate Purkinje cell-evoked IPSPs, making the cell less likely to fire more action potentials, whereas LTD tends to be induced when the postsynaptic activity is limited[52], increasing the probability of postsynaptic spiking.

Some future directions

Although the holy grail of relating cellular phenomena to behavior through the intermediate levels of cerebellar synapses and circuits is not yet at hand, the good news is that it seems possible. Perhaps the greatest gap in our present attempts to build these connections is our level of understanding of the ways in which behaviorally relevant signals are coded. For example, does repeated presentation of a tone CS result in a pattern of mossy fiber activation that will support LTP of mossy fiber-granule cell or parallel fiber–Purkinje cell synapses? Does repeated presentation of a tone CS when combined with an airpuff US at intervals that support associative conditioning result in a pattern of activation that will produce parallel fiber LTD? Does repeated presentation of an airpuff US result in a mode of climbing fiber activation that will produce climbing fiber LTD? As these data become available, cellular electrophysiologists must then make their best efforts to simulate these *in vivo* training signals in reduced preparations such as brain slices and to provide a greater data set for the parameters of these plastic phenomena, particularly in regard to duration.

Another important line of work will involve incorporating a wider range of cerebellar plasticity into computer simulations. Even though the cerebellar network is 'simple' by the standards of the CNS, the proliferation of cellular plastic phenomena has now reached the point where it is not possible to make straightforward predictions about the behavior of the circuit during and after training. Computer models of multiple plastic phenomena in the cerebellar circuit have already given rise to some very unexpected behavioral predictions that have been confirmed experimentally[69]. Moreover, many of the critical experiments to address the behavioral/cellular link will continue to involve the analysis of behavior together with pharmacological and genetic manipulations. Further molecular understanding of the second messengers and substrates involved in these various forms of plasticity will not only constrain certain aspects of these models but will allow for the development of more sophisticated transgenic and pharmacological strategies. Whereas these general approaches are being used in the analysis of memory storage in many brain regions, we believe that the cerebellar circuit offers a unique opportunity to develop a complete and specific theory of the engram.

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